GlucoTwin — Digital Twin Architecture

Fusing static EHR with live wearable + CGM to predict hyperglycemia 60–120 minutes ahead

DATA STREAMS

① STATIC / HISTORICAL — EHR

- Demographics, BMI, ethnicity
- HbA1c, fasting glucose, labs
- Medications (metformin/insulin…)
- Comorbidities (HTN, CKD, retinopathy)
- Family history (genetic-risk proxy)

source: Synthea · MIMIC-IV

② DYNAMIC / REAL-TIME — Wearable + CGM

- Continuous glucose (CGM)
- Heart rate & HRV (RMSSD)
- Steps / activity
- Sleep stages
- Meal events — sampled every 15 min

source: ShanghaiT2DM · PPG-DaLiA · MMASH

FEATURE EXTRACTION

Static feature vector

12 EHR fields, normalized; encodes chronic risk the wearable cannot reveal.

2-hour window features

CGM level/slope/variability, HRV, steps, sleep fraction, recent carbs, time-of-day.

FUSION

LATE FUSION

concatenate

[dynamic window ‖ static EHR]

One fused vector per time step. Both streams required — proven by the ablation study.

TWIN MODEL

HYBRID TWIN MODEL

Gradient-Boosted Trees
deployed predictor · interpretable

+ optional LSTM arm
raw-sequence encoder, same late-fusion head (PyTorch)

Glucose forecaster
+60 min value → Clarke grid

Counterfactual engine
perturb levers → re-score risk

backend: scikit-learn / XGBoost

OUTPUTS

Adverse-event risk
hyperglycemia 60–120 min ahead · ROC-AUC 0.95

Explainability
per-alert SHAP drivers (wearable + EHR)

What-if simulation
walk · meds · sleep · meal → actionable recommendation

CLINICIAN DASHBOARD (Streamlit)

Virtual-patient view · live glucose & risk curve · alert feed · "why at risk" explanations · what-if controls

🔒 **Privacy by design**

Synthetic / open / anonymized data only · DPDP Act (India) & HIPAA compliant · no real patient data · reproducible from a seed